

ANDREW STRATMANN

Overland Park, KS

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EDUCATION

The University of Colorado, Boulder, CO M.S. in Electrical and Computer Engineering	<i>Anticipated May 2027</i> Overall GPA: 3.94/4.0
The University of Kansas, Lawrence, KS M.S. in Computer Science	<i>May 2025</i> Overall GPA: 3.89/4.0
The University of Missouri, Columbia, MO B.S. in Civil Engineering	<i>May 2023</i> Overall GPA: 3.96/4.0
Truman State University, Kirksville, MO B.S. in Computer Science and Mathematics	<i>May 2017</i> Overall GPA: 3.97/4.0

PUBLICATIONS

- A. Stratmann, M. Hashemi, “Efficient Index-Based Multi-User Scheduling for Mobile mmWave Networks: Balancing Channel Quality and User Experience”, *IEEE Consumer Communications & Networking Conference (CCNC)*, 2026.

RESEARCH EXPERIENCE

Efficient Multi-User Scheduling in Millimeter-Wave Networks <i>Thesis Work for M.S. Computer Science</i>	Spring 2024 - Spring 2025 <i>Lawrence, KS</i>
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- Formulated more realistic cost curve for beamforming phase of transmission in 30-300GHz networks
- Utilized new cost curve along with contextual information to intelligently select users for transmission
- Incorporated physically realistic assumptions and evaluated scheduling performance using simulation
- Prepared results in a formal report and for integration with existing user scheduling research

Earlier Research Experiences	2014 - 2016
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- NSF REU in Secure Cloud Computing, Missouri S&T (2016): improvements to Attribute-Based Encryption, implemented in C.
- Summer Mathematics Research Program, Kansas State (2015): extension of the Euler characteristic to higher categories in algebraic topology.
- Summer Institute for Training in Biostatistics, NC State (2014): statistical analysis of medical data in R and SAS, identifying correlations between cardiac events and lifestyle variables.

TEACHING EXPERIENCE

Learning Center, University of Missouri <i>Tutor</i>	August 2021 - May 2023 <i>Columbia, MO</i>
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- Provided assistance to students taking courses in introductory engineering, coding topics, and Calculus I-III
- Completed ancillary trainings to help students at all stages of the learning process

School of Engineering, University of Kansas <i>Graduate Teaching Assistant</i>	August 2017 - May 2019 <i>Lawrence, KS</i>
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- Instructed students in supplementary and review materials from a discrete mathematics course
- Answered questions regarding course materials during office hours and in discussion sections
- Corrected written assignments and exams

Department of Mathematics, Truman State University

Grader/Tutor

2014 - 2017

Kirksville, MO

- Provided assistance to 20 students taking courses in Calculus and Foundations of Mathematics
- Graded 50 homework assignments several times per week from these courses

WORK EXPERIENCE

Garmin International

Software Engineer II - Embedded Avionics

June 2023 - Present

Olathe, KS

- Implemented embedded software solutions to support an aftermarket autopilot product
- Designed and tested calibration solution for an embedded sensor used for airdata
- Collaborated with hardware team to ensure software development makes full use of physical resources
- Defined embedded software requirements and ensured adequate testing thereof
- Prepared integrated avionics software for DO-178B certification process

Cerner Corporation

Software Engineer I

June 2019 - August 2021

Kansas City, MO

- Led data syndication platform migration for 10+ hospital systems by implementing conversion libraries and authoring migration docs enabling deprecation of the legacy platform
- Implemented and maintained API services to support Healthcare IT solutions
- Managed 20+ monthly production deployments to support client hospitals and government institutions
- Mentored 4 new-hires in areas of technical onboarding and software development competencies

TECHNICAL STRENGTHS

Languages

C (embedded), Python (NumPy, SciPy), MATLAB, Ruby, Java

Hardware & Embedded

JTAG debugging (Lauterbach), Verilog, VHDL

Wireless/RF

SDR (introductory), FCC licensed amateur radio (KF0CHO)

Research Tools

Git, L^AT_EX, Docker

AWARDS AND ACCOLADES

Bonnie Zelenak Excellence in Tutoring Award, 2023

University of Missouri

Outstanding Undergraduate in Computer Science, 2017*

Truman State University

Outstanding Undergraduate in Mathematics, 2017*

Truman State University

**Awarded by each academic department at Truman State University once per year to a single undergraduate student*